Q1. Write short notes on fixed and variable addresses. Explain the memory-mapped I/O concept using a scenario where port # 05h is connected with a keyboard and port # 02h is connected with a 7-segment display. 3+4

Q2. What are the main reasons to use microcontroller? What should be a mega microcontroller's name if it is manufactured by ATMEL along with 256 Kbyte of flash memory? 3+2

Q3. What is Harvard Architecture for microcontroller design? Explain the use and functions of appropriate pins of ATMega16 microcontroller for analog I/O and external clock input operation. 1+2

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Q1. How do the Daisy-Chain and Multi-level Bus architectures differ from each other? Explain. 05

Q2. If an Interrupt is given serviced at a Master 8259 for 7th Slave 8259, then how can the Master 8259 notify it to the Processor? Explain with example. 05

Q3. Draw the control word format for the 8255A PPI (*Consider that Port-A is in Mode-1 as an input port, Port-B is in Mode-0 as an output port and the lower-nibble of Port-C is working as an input and upper-nibble of Port-C is working as ~~an output~~ handshaking mode*). 05

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